Mathematical overview

Model and first-visit facts. Use a finite connected undirected map, whole printed links and strictly positive prices. Buying pays once per link; traveling pays for every traversal. Length on paper is not price. With every place required, a cheapest network is a tree, with $n-1$ links on n places: a loop could lose a link and cost less. The first visit's other useful fact is the exchange certificate: a tree is cheapest exactly when no unused link is cheaper than a link on the tree route between its ends.

Cheapest networks solve every lowest-dearest-route problem at once. A route's *dearest* link is its largest price, not its total. Every route along a cheapest all-place network has the smallest possible dearest price among routes with those endpoints. If a cheaper-dearest route existed, remove a dearest tree link. That separates the tree into two sides; the proposed route crosses the separation on a cheaper link, giving a cheaper tree. Conversely, a tree with this property for *every* pair is cheapest, by the exchange certificate. Merely minimizing the largest price anywhere in the network is not enough.

Opening links determines the price list and cost. Open *all* links priced at most k, not just a chosen network. The reachable groups only merge. Two places first share a group at exactly their lowest possible dearest price. Let c_k be the number of groups and $c_0 = n$. Every cheapest tree has exactly $n - c_k$ links priced at most k: its low-price pieces must join every group, by the preceding route fact, and each piece is a tree. Thus it has $c_{k-1} - c_k$ links of price k. All cheapest trees share one price list even when their links differ. For positive *integer* prices,

$$\text{cheapest cost} = \sum_{k\geq 1} k(c_{k-1} - c_k) = \sum_{k\geq 0} (c_k - 1).$$

The sums stop when there is one group. Include thresholds at which nothing merges. This is an explanation of children's records, not a buying recipe to give them first.

Trips and networks react differently to repricing. Adding a to every price adds $a(n-1)$ to every all-place tree, so it changes no cheapest network while prices stay positive. A trip of j links gains aj, so different-length trips can overtake. Doubling preserves both kinds of optimum. More generally, strictly increasing repricing preserves all cheapest all-place trees, including ties, because it preserves every exchange comparison. Squaring positive prices can nevertheless change cheapest trips.

Optional places change the problem. Circles must connect; squares need not. A cheapest purchase has no loop and no optional leaf: either could be trimmed for a saving. With two required places it is a cheapest trip; with all places required it is the original tree problem. In between, a purchase can resist every cheaper single change yet not be cheapest. With $r \geq 2$ required places, at most $r-2$ optional places can have three or more bought links; the leaf-count proof is on guide p. 8. The preceding all-place price-list, repricing and converse claims do *not* extend to optional places.

Discovery and evidence. Building supplies examples; a failed search is not impossibility. A complete small catalog proves that case; the cut and leaf arguments explain general claims. K–1 compares reachable groups and optional-place payment piles (fixed maps: dots to 4, totals to 12). Grades 2–3 compares dearest links, sums and invented prices (printed comparisons to 18). Grades 4–5 connects catalogs to proofs and tests where local improvement fails. Reading and recording may be adult-supported; link choices stay with children.

Preparation and shared launch

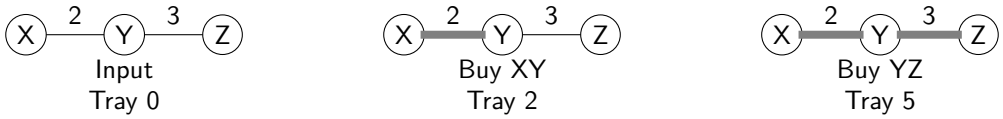
Status. Unpiloted and physically unrehearsed. Print scale, tape clearance, pawn fit, counter handling, preparation time and the proposed pacing still need a real rehearsal. Digital checks do not establish classroom readiness.

Kit per pair or upper trio. Reuse 30 equal counters about 12–15 mm across, ten removable tape strips about 5×20 mm, one small pawn or a finger, one sleeve and dry-erase pen, an off-map tray, and one sheet of scrap paper as a second payment area. The young and middle tables each have two kits (60 counters, 20 strips, two of each other item); the upper trio has one. Add 15 counters and four strips for demonstration: 165 counters and 54 strips overall. These are planning quantities for five working groups, not an attendance claim.

Print and prepare. US Letter, single-sided, 100%. The return-visit files are `k-1.pdf` (7 pages), `grades-2-3.pdf` (6), and `grades-4-5.pdf` (10). Print selected pages once per pair/trio, not one full packet per child. The large maps are working mats; small duplicate maps and examples are pen-only records. Working place circles are 10 mm across, with centers roughly 40 mm or more apart. No pages are cut. Base Week 53 has *no materials PDF*; no base student page is needed for the routes below. For an optional familiar comparison, print `week-53-students.pdf` p. 3 (trap) or p. 4 (five-place map) as an uncut mat. Current base guide: W53-guide-v4.

Allow a proposed 15–20 minutes to print selected mats, count kits and prepare strips; this estimate is unmeasured. Before class, place strips on the tightest map you will use. Keep price labels visible and counters off the roads. If a pawn hides a place, use a finger. Thirty counters cover the listed optimal single payments, *not every possible purchase*: the squared-price map even has a price-36 link. Write large trial totals rather than demand an oversized payment pile. On the six-place comparisons, $18 + 15 = 33$ and $16 + 16 = 32$ exceed one kit: write the first total, return its counters, then pay the second.

Whole group, about 5 minutes after free handling. Draw this non-task line on the board; circles about 10 mm wide and centers at least 40 mm apart. Let a child choose and mark one whole link; show its payment, then finish the line. This is a convention example, not an optimization question.



Trace X–Y–Z again: a bought network owes nothing more. Show returning a link and its refund. Introduce the next new action *at its table*, using the printed start/step/result example: opening links (K–1 p. 2; middle p. 2; upper p. 3), trips (middle p. 3; upper p. 6), or squares (K–1 p. 4; middle p. 6; upper p. 8). For trips, record one word, then pay: on this line XYZ costs 5; XYZY costs 8. For squares, a pawn may pass through the square but need not visit it. Do not introduce three new objectives in one speech.

Staffing and one hour. Anchor one adult at each table; read or scribe without choosing links. In the trio rotate chooser, payer/checker and recorder. Use 5 minutes movement, 5 handling/launch, 20 exploring, 5 break, 20 continuing and 5 sharing. Young first route: P1–3; later P4–6, with P7–8 readiness-dependent. Middle: P1–3, then P4 if ready; save P5–8. Upper: P1–4, then one of P5–6; later P7–10, with P11–12 for proof-ready children. Finishing every page is not the goal.

K–1 solutions and discretionary hints

AB means the whole link; buying it first or last does not make a new network. Spare record maps are workspace, not a promise of that many answers. Let children choose; the adult can read, count and draw their chosen links.

Problem 1 page 1

Minimum **4**, exactly three networks: {AB, AC, CD}, {AB, BC, CD}, {AC, BC, CD}. D needs a link costing at least 2; two more links cost at least 1 each. Equality uses CD and any two sides of the A–B–C triangle. All three work, so the list is complete. *Hint if needed.* "Can the pawn reach D? Which road could it use?" If only one answer appears, ask whether the same payment pile can buy a different pair in the triangle.

Problem 2 page 2

With 1-dot links open, groups are **AB / C / D / EF**; up to 2 dots, **ABC / DEF**; up to 3 dots, **ABCDEF**; up to 4 dots, still one group. The pawn first reaches F from A at **3**. Include isolated places in rings. Opening means marking every eligible link; no payment total is requested. *Hint if needed.* "Where can the pawn go without crossing an unmarked road?" A group need not look round; its membership matters.

Problem 3 page 3

Minimum **9**, in two ways: {AB, AC, CD, DE, EF} and {AB, BC, CD, DE, EF}. The cheapest connecting price list is $1, 1, 2, 2, 3$: AB and EF are forced by their cheapest exits, DE joins D to E cheaply, and either AC or BC connects C. CD is the cheapest link between ABC and DEF. No network can avoid all 3- and 4-dot links, since the 1- and 2-dot links leave those two groups separate. *Hint if needed.* "Can you cross from the left ring to the right ring without a 3 or 4?"

Problems 4 and 5 pages 4 and 5

Each outer triangle has three links priced 4. Without S, any two cost **8**, the minimum. For P4, using S costs **6**, uniquely AS, BS, CS. For P5's top map, using S costs **9**, uniquely the three spokes: the square costs one extra. On P5's bottom map, using S costs **8**: AS, BS together with *one of* CS, AC or BC. The square ties rather than saves. Those are all three cheapest ways using S; adding the three outer pairs gives six cheapest overall.

Why complete: a cheapest join using S is a tree. It uses three spokes, two spokes and one outer link, or one spoke and two outer links. Comparing these three forms gives the stated minima. The last form is wasteful when S is optional, but it must not be silently forbidden in a comparison that requires using S. *Hint if needed.* P4: "Do all three circles have to use an outside road?" P5: "Must using the square mean buying all its roads?" Keep both piles for these small comparisons.

Problem 6 page 6

Without X: **12**, any three outer sides (four choices). Using X: **8**, uniquely AX, BX, CX, DX, saving 4. Without X, a four-place tree needs three price-4 links. With X, a spanning tree on five places needs four links, each at least 2; four spokes attain that bound. *Hint if needed.* "How many circles can your marked roads reach?" If ready, ask whether any other four-link purchase can still cost 8.

Young continuation and middle entry

K–1 Problem 7 page 7

Minimum **6**, uniquely AS, BS, ST, CT, DT. Without either square the cost is 10 (AB, AC, CD); allowing only S or only T lowers it to 9. With both, five links are needed and only four price-1 links exist, so cost is at least $1+1+1+1+2=6$. *Hint if needed.* "What would join the two small groups through their squares?" This is a continuation, not a required young-table proof.

K–1 Problem 8 page 7

Use positive whole-number dots. A working answer is **2, 2, 4** on the three spokes; 1, 3, 4 and 2, 3, 3 also work, in any arrangement. Complete rule: sort the spoke prices $x \le y \le z$. The cheapest join using S costs $\min(x+y+z, x+y+4, x+8)$. It equals 8 exactly for $(1,3,z)$ **or** $(2,2,z)$ **with** $z \ge 4$**, or** $(2,3,3)$. Do not cap invented dots unless reducing the task deliberately. *Hint if needed.* "Could two square roads and one outside road beat your three roads?" A sum of 8 alone fails: 1, 1, 6 has a cost-6 alternative. The printed prompt leaves positivity implicit; say "at least one dot per road" before inventing prices.

Grades 2–3 solutions and discretionary hints

Problem 1 page 1

Minimum **9**. Every cheapest network uses CD and EF, any two of AB, AC, BC, and one of CE, DE. The complete six are:

AB, AC, CD, CE, EF	AB, AC, CD, DE, EF
AB, BC, CD, CE, EF	AB, BC, CD, DE, EF
AC, BC, CD, CE, EF	AC, BC, CD, DE, EF

The three price-1 links chosen must join ABC and EF; CD is D's uniquely cheapest exit. A price-4 link is still needed between ABCD and EF, with two choices. The lower bound $1+1+1+2+4=9$ is attained in exactly those 3×2 ways. *Hint if needed.* "Which part is still separate? Can two pairs share different triangle choices?"

Problems 2 and 3 pages 1 and 2

Pair	One lowest-dearest route	Dearest price	First common ring
A to D	ACD	2	2
B to D	BCD	2	2
D to F	DEF	4	4
A to F	ACEF	4	4

Other routes can be equally good; these problems do not request all routes. For P3: at price 1 the groups are **ABC / D / EF**; at 2, **ABCD / EF**; at 3, unchanged; at 4, one group. D is isolated below 2, and EF is separate below 4, proving the lower bounds. *Hint if needed.* P2: "Are we adding the prices or choosing the biggest?" P3: "Did the price-3 road actually join different groups?"

Grades 2–3 trips and optional places

Problem 4 page 3

All three cheapest networks are AB, BC, CD. Their costs are **3, 6, 9** for original, $+1$ and $+2$ prices. Cheapest A–D trips are respectively **ABCD at 3; AD at 5; AD at 6**, each unique. These are the only two simple A–D routes; a repeated part costs extra. *Hint if needed.* "Which roads stayed the same even though their total changed?" Record links as well as costs, using the printed lines.

Problem 5 page 4

All three cheapest networks are **AB, BC, CD, DE, EF**. Complete trip key:

Prices	Network cost	Every cheapest A–F trip	Trip cost
Original	8	ABCDEF, ACDEF	8 each
Every price 2 more	18	ACDF	15
Doubled	16	ABCDEF, ACDEF	16 each

For the network, AB, BC, CD, DE, EF are uniquely cheapest across cuts A / rest, AB / rest, ABC / DEF, ABCD / EF and ABCDE / F respectively. For trips, both original ways to reach C cost 3; then CDE costs 3 and EF costs 2. After $+2$, AC wins to C and DF wins from D to F. Guide p. 7 gives a complete shortest-trip certificate. *Hint if needed.* "Can a road we did not buy still be used on a trip?" Yes: a trip is on the whole printed map. Reuse counters between the 18/15 and 16/16 comparisons.

Problems 6 and 7 page 5

P6: write AB$=a$, BC$=b$, AC$=c$, using positive whole numbers. For a *strict* switch, $a+b<c<a+b+2$, hence $c=a+b+1$. For example 1, 1, 3; 1, 2, 4; 2, 2, 5. The two-link trip gains 4 while the direct trip gains 2. *Hint if needed.* "How much did each trip gain?" If a child allows tied cheapest trips, $a+b\le c\le a+b+2$ also meets a weak reading of the prompt; ask whether they intended a tie or a clear switch, rather than rejecting it.

P7: **No; no**, for the no-square maps used so far and changes keeping prices positive. Every connecting tree has the same number of links, so adding the same amount preserves its ranking; doubling doubles every trip total. *Hint if needed.* "Do all networks gain the same number of extra counters? Do all trips?" P8 changes the rules, so the first answer need not survive there.

Problem 8 page 6

Left: **8 without S, 6 using S**. Right: **8 without, 9 using**. Without, choose any two outer links; using, the unique minimum is AS, BS, CS on each map. The three tree forms on guide p. 3 prove the comparisons. *Hint if needed.* "Is an optional place always a bargain?" Let the child demonstrate the two piles before discussing a general rule.

Grades 4–5 routes groups and general claims

Problem 1 page 1

The answers are **A–D 2 via ACD; B–D 2 via BCD; D–F 4 via DEF; A–F 4 via ACEF**. Other routes are valid. D cannot be reached below 2; crossing between ABCD and EF requires at least 4. *Hint if needed.* "Which groups would remain apart if that largest price were forbidden?"

Problem 2 page 2

Minimum **9**, exactly the six networks listed under middle P1 on guide p. 4. In each choose two of AB, AC, BC, then CD, EF and one of CE, DE. The product of three triangle choices and two bridge choices proves completeness; buying order is irrelevant. *Hint if needed.* "Can you organize the records by what changes near A, then by what changes near E?" Pool the six cases across the trio.

Problem 3 page 3

At thresholds 1, 2, 3, 4 the groups are **ABC / D / EF; ABCD / EF; ABCD / EF; ABCDEF**. Starting with no open links gives six groups, so counts $c_0, \ldots, c_4$ are **6, 3, 2, 2, 1**. The first common thresholds for A–D, B–D, D–F, A–F are **2, 2, 4, 4**. Opening the price-3 link changes no group. *Hint if needed.* "If a road opens inside one ring, does the number of groups change?" Isolated places still count as groups.

Problem 4 page 4

Minimum **6**, exactly three networks:

$$\{\text{AB, AC, CD, DE}\} \qquad \{\text{AB, BC, CD, DE}\} \qquad \{\text{AC, BC, CD, DE}\}.$$

At price 1 the groups are **ABC / D / E**; at price 2 all five places join, and later thresholds do not change that. All three networks have prices **1, 1, 2, 2**. A five-place tree needs four links. Only two price-1 triangle links can be used without a loop, so cost is at least $1 + 1 + 2 + 2 = 6$. CD and DE are the only price-2 links; the three triangle pairs complete the list. *Hint if needed.* "Can three price-1 links help reach D or E?"

Problem 5 page 5

Sam is wrong. Every loop-free connecting network on the square uses three sides and has largest price 3. The network BC, CD, AD costs **9**; each of the other three costs **7**, using AB and any two of BC, CD, AD. Largest price 3 is unavoidable but does not determine total cost. *Hint if needed.* "Can two networks have the same dearest road and different payment piles?" Distinguish Sam's one number for the whole network from P6's condition for every pair.

Problem 6 page 5

Yes; no; yes, on a connected positive-price map with no squares. Every cheapest-network route is lowest-dearest; all cheapest networks have the same price list; threshold groups determine cost. The proof ideas open this guide. Two examples suggest, but do not prove, a universal statement.

For P2–3 the group drops are $6 - 3 = 3$, $3 - 2 = 1$, $2 - 2 = 0$, $2 - 1 = 1$. Therefore there are **three price-1 links, one price-2 and one price-4**, giving $3 + 2 + 4 = 9$. Alternatively add $5 + 2 + 1 + 1 = 9$. For P4, the drops are $5 - 3 = 2$, $3 - 1 = 2$, giving $2 + 4 = 6$, or $(5 - 1) + (3 - 1) = 6$. Count the threshold-3 pause in the first map even though no link of that price is bought.

Hint if needed. "Suppose a cheaper-dearest route existed. What happens if we lift the dearest road in your tree?" Draw the two resulting sides and let the child find where the proposed route crosses. For the price-list question ask, "How many links are needed inside each ring?" Save the converse for P12.

Limit. P9's 2,2,4-spoke map has cheapest lists 4,4 and 2,2,4: optional squares change the theorem.

Grades 4–5 trips and optional places

Problem 7 page 6

Networks are **AB, BC, CD** in all three maps, costing **3, 6, 9**. Cheapest trips are **ABCD at 3; AD at 5; AD at 6**. Each is unique: the two simple routes compare 3 with 4, 6 with 5, and 9 with 6. *Hint if needed.* "Which payment counts every bought road, and which counts only your trip?" Reuse the printed trip example before comparing totals.

Problem 8 page 7

Every cheapest network is **AB, BC, CD, DE, EF**, uniquely. The cut argument under middle P5 proves this, and strictly increasing changes preserve all its strict comparisons.

Prices	Network cost	Every cheapest A–F trip	Trip cost
Original	8	ABCDEF, ACDEF	8 each
Every price 2 more	18	ACDF	15
Every price doubled	16	ABCDEF, ACDEF	16 each
Each price times itself	14	ABCDEF	14

Complete trip certificate. The least costs from A to A,B,C,D,E,F are respectively

	A	B	C	D	E	F
original	0	1	3	5	6	8
$+2$	0	3	5	9	12	15
$\times 2$	0	2	6	10	12	16
squared	0	1	5	9	10	14

Every printed link satisfies $d(v) \le d(u) + w(uv)$ in both directions; summing along a route proves its cost is at least the destination label. The listed trips attain the bound. Equality requires each step to advance by exactly its price: originally C can follow either A or B, while D, E, F must follow C, D, E; after $+2$ the successful chain is A,C,D,F; doubling keeps the original choices; squaring forces A,B,C,D,E,F. Thus no cheapest trip is missing. These adult labels are not an extra child ledger.

None of the three changes alters cheapest *all-place* networks. Addition and squaring can change cheapest trips; doubling cannot. *Hint if needed.* "What does adding 2 do to a three-road trip compared with a five-road trip?" For squaring ask which costly single road now competes with several cheaper ones. Never ask the kit to pay the squared price 36; numerical comparison is enough.

Problem 9 page 8

Left star: cost 8, three ways, AB/AC, AB/BC, AC/BC; S is unused. A tree using S costs at least 9. **Right star: cost 8, six ways**: those same three outer pairs, plus {AS, BS, CS}, {AS, BS, AC}, {AS, BS, BC}. The three tree forms from guide p. 3 show completeness. **Bottom map: cost 8, two ways**, {AB, BC, CD, DE, EF} and {AC, CD, DE, EF}. Only A and F are circles; B,C,D,E are squares. A positive-price optimum for two required endpoints is just a cheapest simple trip, so P8's original trip list proves completeness. *Hint if needed.* "Which places must be reached here? Could you return a road that goes only to an optional dead end?"

Problem 10 page 9

Cheapest is **6**, uniquely AS, BS, CS. From AB and BC, the answer is **no**: with a cap of 8, a purchase with no spoke can only be an outer pair. Returning a link disconnects a circle; buying any link exceeds 8; swapping an outer link for one spoke still disconnects a circle. Only swaps to the other outer pairs survive. The same holds with cap 9. A cap of **10** is enough: buy AS (10), swap AB for BS (8), swap BC for CS (6). *Hint if needed.* "Could one new spoke connect both sides left by returning an outside road?" A swap is one atomic exchange; its final state must connect the circles. This optional-place example, not a contradiction, marks the limit of the first-visit exchange test.

Upper continuations explanations and sources

Problem 11 page 10

Minimum **6**, uniquely AS, BS, ST, CT, DT, as in young P7 (guide p. 4). A square place cannot have just one bought link in an unrestricted cheapest purchase: that link is a positive-price dead end, removable without separating any required circle. A square can be unused (degree zero); do not confuse that with an optional leaf in the bought tree.

With four circles, the largest possible number of squares with three or more bought links is **2**. This map attains it at S and T. Here is a proof that also covers larger maps. Ignore unused places. A cheapest purchase is a tree; every leaf is required. If there are L leaves and the branch places have degrees $d_1, \ldots, d_b \geq 3$, counting link ends gives

$$-L + \sum_{i=1}^{b}(d_i - 2) = -2, \qquad L = 2 + \sum_{i=1}^{b}(d_i - 2).$$

Degree-2 places add nothing. Each branch place accounts for at least one extra leaf beyond two, so there are at most $L - 2$ branch places altogether. Since $L \leq 4$, at most two can be optional branch places. More generally r required places give at most $r - 2$ such squares, for $r \geq 2$.

Hint if needed. "What happens to the payment if we return the only road to that square?" For the bound, ask, "When a path splits again, how many extra ends must eventually appear?" Children may explain with drawn branches; algebra is adult backup. Two branches on this particular map show attainability, not by themselves a universal bound.

Problem 12 page 10

Lee is right, for a connected map with no squares and positive prices. His condition implies every unused link f costs at least every link on the candidate tree route between its ends: f itself is an available route. This is the exchange certificate. The following transformation proves its sufficiency without assuming a buying algorithm.

Let T be Lee's tree and R any competing spanning tree. Choose a link e of T missing from R. Deleting e splits T into two sides. Adding e to R makes a loop; some other loop link f crosses those sides. The only T link across that split was e, so f is not in T. The T route between f's ends uses e, and Lee's condition gives $w(e) \leq w(f)$. Replace f with e in R: cost cannot rise, and agreement with T increases. Repeat until R becomes T. Thus T costs no more than any spanning tree; any connected purchase with loops can first shed positive-price links. T is globally cheapest.

Hint if needed. "What does Lee's condition say about a route consisting of just one unused road?" If the full transformation is too abstract, enact one exchange, retain the conjecture, and save the proof. Sam's claim in P5 checks only the whole network's largest price; Lee checks every pair, a much stronger condition.

Sources consulted and boundaries

Consulted: the final three student PDFs (W53-RV-K/M/U-v1), editable sources, earlier page reviews, return-visit README and `week-53-return-visit.md` outline; base Week 53 guide, especially pp. 1–2 and its exchange argument; Week 8 return-visit and Week 98 model guides. Old review numbering predates the revised pages; this guide and its independent checker use the final PDFs.

The outline cites T. C. Hu, "The Maximum Capacity Route Problem," *Operations Research* 9(6), 1961, pp. 898–900 (only its abstract checked by the outline author); Sedgewick and Wayne, *Algorithms*, 4th ed., §4.3 companion, Exercise 1 and Web Exercise 1 (repricing and bottleneck distinction); and the Wikipedia articles "Widest path problem" and "Steiner tree problem." Hwang, Richards and Winter, *The Steiner Tree Problem*, 1992, is background cited there, not consulted here. Those originals were not reconsulted; this guide supplies elementary proofs and independent finite checks.

Base-week research notes give the activity lineage: CS Unplugged Activity 9, "The Muddy City," printed pp. 76–80, PDF pp. 1–5, and *Math Circle by the Bay*, "How we teach," printed p. ix/PDF p. 10. The notes, not those unavailable originals, were consulted. Materials and pacing are untested proposals. Record pages used, handling problems and adult decisions needed.